

SUNIL KIRAN NAYAK

Software Engineer | Python Backend Developer | Django | FastAPI | AWS

+91 7894140775 | sunilkirannayak@gmail.com | <https://www.linkedin.com/in/sunil-kiran-nayak-820b95289/> | <https://github.com/sunilhithuhb>

PROFESSIONAL SUMMARY

Backend Developer (3+ yrs) building scalable Python/FastAPI/Django systems for production and AI-powered platforms — from multi-tenant SaaS to LLM-powered document search.

Key Highlights

- Cut average API response time by 63% through database replication and query tuning.
- Cut system downtime by 40% by architecting a proactive error-logging and alerting pipeline.
- Reduced dashboard load time by 60% and query speed by 30% on a 50+ client SaaS analytics platform.

Core Skills: Python, Django, FastAPI, REST API Development, PostgreSQL, MySQL, AWS, Docker, CI/CD, Redis, Celery, System Design, Database Optimization, Agile Development

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, DRF, FastAPI, SQLAlchemy, REST API Design

AI/LLM Tooling: LangChain, LangGraph, Amazon Bedrock, Amazon Kendra, RAG

Databases & Caching: PostgreSQL, MySQL, Redis, Celery, Replication, Sharding, Query Optimization

Cloud & DevOps: AWS (EC2, ECS, S3, Transcribe, Polly, Bedrock, Kendra), Docker, GitHub Actions (CI/CD), boto3

Security: JWT, RBAC, OTP Verification, Input Validation, bcrypt

Tools & Integrations: Git, Postman, Swagger/OpenAPI, JIRA, Azure Face API, Firebase, Pandas

PROFESSIONAL EXPERIENCE

Software Engineer — *[Madul Tech Software Private Limited]*

2023 – Present

- Architected a proactive error-logging and alerting pipeline, cutting system downtime by 40%.
- Automated database archival to cold storage, improving query performance by 25% and cutting storage costs by 15%.
- Unified REST API contracts across web and mobile clients, boosting feature adoption by 25% and reducing support tickets by 15%.
- Introduced feature-flagging and modular API versioning, accelerating release cadence by 20%.
- Built CI/CD pipelines (GitHub Actions + Docker) across dev and production, enabling zero-downtime releases.
- Led backend and frontend engineers through task planning and milestone tracking, improving delivery efficiency by 20%.
- Optimized legacy storage and search queries through database replication and tuning, cutting average API response time by 63%.
- Translated technical requirements into scalable API specifications, improving development efficiency by 17%.
- Delivered 2 full-stack projects end-to-end using Python, AWS EC2, PostgreSQL.
- Implemented database sharding to scale high-traffic workloads, improving query response times and throughput.
- Built automated exception detection and alerting, cutting mean-time-to-resolution (MTTR).
- Diagnosed and resolved 3 major production anomalies, restoring stability and preventing recurrence.
- Built bulk email/SMS notification pipelines integrating third-party APIs, reducing delivery latency at scale.
- Partnered with frontend engineers on API contract design, cutting round-trip debugging cycles.

PROJECTS

FastAPI Document Intelligence API

Tech Stack: Python, FastAPI, LangChain, Amazon Bedrock, Amazon Kendra, PostgreSQL, Docker

- Rebuilt a document search product backend in FastAPI, exposing RESTful endpoints for document ingestion, semantic search, and Q&A powered by LangChain's RetrievalQAChain with Amazon Bedrock (LLM) and Kendra (vector retriever).
- Implemented company-wise metadata isolation using PostgreSQL row-level scoping, ensuring multi-tenant data security with zero cross-tenant leakage.

- Added speech-to-text and text-to-speech endpoints using AWS Transcribe and Polly, reducing information retrieval time and improving accessibility.
- Containerized the service with Docker and deployed on AWS ECS behind an Application Load Balancer; reduced cold-start latency by 30%.

TrendTrax — SaaS Digital Marketing Analytics Platform

Tech Stack: *Python, Django, DRF, LangGraph, PostgreSQL, Redis, Celery, AWS, Pandas*

- Built a scalable multi-tenant analytics platform using Django, supporting 50+ clients with secure data isolation.
- Designed REST APIs to ingest and process 100K+ records/day from Google Analytics & Ads APIs.
- Implemented async pipelines using Celery & Redis, reducing API latency by 60%.
- Integrated LangGraph workflows for automated insights, improving data processing efficiency by 40%.
- Optimized performance with Redis caching and PostgreSQL tuning, improving dashboard load time by 60% and query speed by 30%.

HRMS — Human Resource Management System

Tech Stack: *Python, Django, MySQL, AWS S3, Azure Face API, Firebase, Celery*

- Designed and implemented a comprehensive employee management backend with RESTful APIs covering employee profiles, attendance tracking, leave management, and announcements — built using Django with Django ORM over MySQL/PostgreSQL.
- Integrated Azure Face API for automated facial recognition during employee login/logout, enabling touchless biometric attendance tracking with high accuracy.
- Built a Leave Management System (LMS) with role-based approval workflows, allowing employees to apply, track, and manage leave requests through structured API endpoints.
- Implemented a Celery cron job to automatically generate and dispatch birthday and work-anniversary announcements via email and in-app notifications, reducing manual HR effort.
- Built an async AWS S3 service layer in Python for secure document uploads (employee records, payslips), leveraging boto3 with pre-signed URLs for access control.
- Collaborated with frontend developers on API integration, cutting integration bugs through clean contract definitions.

EDUCATION

Master of Computer Application (MCA) NM institute of Engineering and Technology, Odisha 78.00%	2022-2024
Bachelor of Science (B.Sc) Sailendra Narayan college, Odisha 79.00%	2019-2022